capable of handling less than 0.1 percent of the throughput. Ethereum's lack of scalability places DeFi at risk of being unable to meet requisite demand. Much effort is focused on increasing Ethereum's scalability or replacing Ethereum with an alternative blockchain that can more readily handle higher transaction volumes. To date, Ethereum's long awaited version two has not been implemented. However, some new platforms such as Polkadot, Zilliqa, and Algorand offer some solutions for this scaling risk.[16]

One actively pursued solution to the problem is a new consensus algorithm, *proof of stake,* which is introduced in Chapter 3. Proof of stake simply replaces mining of blocks (which requires a probabilistic wait time), with staking an asset on the next block, with majority rules similar to proof of work. *Staking*, an important concept in cryptocurrencies and DeFi, means a user escrows funds in a smart contract and is subject to a penalty (*slashed funds*) if they deviate from expected behavior.

Malicious behavior in proof of stake occurs with voting for multiple candidate blocks. This action shows a lack of discernment and skews voting numbers; thus, it is penalized. The security in proof of stake is based on the concept that a malicious actor would have to amass more of the staked asset (ether in the case of Ethereum) than the entire rest of the stakers on that chain. This goal is infeasible and hence results in strong security properties similar to proof of work.

Vertical and horizontal scaling are two additional general approaches to increasing blockchain throughput. Vertical